# Examen
## Data Dictionary

Server: MySQL Local
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# 1. Introduction

A data dictionary is a collection of metadata such as object name, data type, size, classification, and relationships with other data assets. It is used by data administrators, analysts, and engineers to understand and trust data assets. It helps in the creation of authentic, transparent, and consistent data throughout the organization.

In this data dictionary, following servers are chosen for documentation:
MySQL Local (MySQL)

For each server, definitions of tables (entities/collections), views, functions are listed in order. Detailed properties of data elements (data type, size, nullability, optionality, indexes, foreign keys, constraints) are organized in tabular format.

# 2. Server: MySQL Local (MySQL)

Version: 8.0.44

1 database is listed below.

## 2.1. Database: ecommerce_230173

Database objects to be listed: 17 tables, 13 views, 27 functions

### 2.1.1. Tables

#### 2.1.1.1. Table: bitacora

Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |
| 2 | tabla | varchar(100) | ✓ | |
| 3 | operacion | enum | ✓ | |
| 4 | usuario | varchar(100) | ✓ | |
| 5 | descripcion | text | | |
| 6 | fecha_hora | datetime | | Default: CURRENT_TIMESTAMP |

#### 2.1.1.2. Table: carrito

Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |
| 2 | sesion_id | int | | Index: fk_sesion_idx<br>idx_carritos_sesion |
| 3 | divisa_id | int | ✓ | Default: 1<br>Index: FK_divisa_idx |
| 4 | total_productos | int | ✓ | Default: 0 |

| | | | | |
|---|---|---|---|---|
| 5 | monto_aproximado | double | ✓ | Default: 0 |
| 6 | fecha_creacion | datetime | ✓ | Default: CURRENT_TIMESTAMP<br>Index: idx_carritos_fecha |
| 7 | fecha_actualizacion | datetime | | |
| 8 | estatus | enum | ✓ | Default: 'Activo' |

## Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| FK_divisa_idx | NORMAL | BTREE | divisa_id |
| fk_sesion_idx | NORMAL | BTREE | sesion_id |
| idx_carritos_sesion | NORMAL | BTREE | sesion_id |
| idx_carritos_fecha | NORMAL | BTREE | fecha_creacion |

## Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_divisa | divisa_id | ecommerce_230173.divisas | id | RESTRICT | RESTRICT |
| fk_sesion | sesion_id | ecommerce_230173.sesiones | id | RESTRICT | RESTRICT |

## Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| carrito_AFTER_INSERT | AFTER | ✓ | | |
| carrito_AFTER_UPDATE | AFTER | | ✓ | |
| carrito_AFTER_DELETE | AFTER | | | ✓ |

### 2.1.1.3. Table: carrito_detalle

## Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | carrito_id 🔑 | int | ✓ | Index: fk_carrito_idx idx_detalle_carrito idx_detalle_carrito2 |
| 2 | producto_id 🔑 | int | ✓ | Index: fk_producto2_idx idx_detalle_producto |
| 3 | cantidad | int | ✓ | |
| 4 | precio_unitario | double | ✓ | |
| 5 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |
| 6 | fecha_actualizacion | datetime | | |
| 7 | estatus | enum | ✓ | Default: 'Activo' |

## Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_carrito_idx | NORMAL | BTREE | carrito_id |
| fk_producto2_idx | NORMAL | BTREE | producto_id |
| idx_detalle_carrito | NORMAL | BTREE | carrito_id |
| idx_detalle_carrito2 | NORMAL | BTREE | carrito_id |
| idx_detalle_producto | NORMAL | BTREE | producto_id |

## Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_carrito | carrito_id | ecommerce_230173.carrito | id | RESTRICT | RESTRICT |
| fk_producto2 | producto_id | ecommerce_230173.productos | id | RESTRICT | RESTRICT |

## Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| carrito_detalle_AFTER_INSERT | AFTER | ✓ | | |
| carrito_detalle_AFTER_UPDATE | AFTER | | ✓ | |
| carrito_detalle_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.4. Table: categorias

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |
| 2 | nombre | varchar(100) | ✓ | |
| 3 | descripcion | text | | |
| 4 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |
| 5 | estatus | enum | ✓ | Default: 'Activa' |
| 6 | total_productos | int | ✓ | Default: 0 |
| 7 | categoria_padre_id | int | | Index: fk_categoria_padre |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_categoria_padre | NORMAL | BTREE | categoria_padre_id |

### Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_categoria_padre | categoria_padre_id | ecommerce_230173.categorias | id | RESTRICT | RESTRICT |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| categorias_AFTER_INSERT | AFTER | ✓ | | |
| categorias_AFTER_UPDATE | AFTER | | ✓ | |
| categorias_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.5. Table: clientes

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |

| | | | | |
|---|---|---|---|---|
| 2 | persona_id | int | ✓ | Index: persona_id_UNIQUE fk_persona |
| 3 | tipo_frecuencia | enum | ✓ | Default: 'No definido' |
| 4 | monto_acumulado_compras | double | | Default: 0 |
| 5 | estatus | enum | ✓ | Default: 'Activo' |
| 6 | fecha_registro | datetime | | Default: CURRENT_TIMESTAMP |

## Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| persona_id_UNIQUE | UNIQUE | BTREE | persona_id |
| fk_persona | NORMAL | BTREE | persona_id |

## Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_persona | persona_id | ecommerce_230173.personas | id | RESTRICT | RESTRICT |

## Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| clientes_AFTER_INSERT | AFTER | ✓ | | |
| clientes_AFTER_UPDATE | AFTER | | ✓ | |
| clientes_AFTER_DELETE | AFTER | | | ✓ |

# 2.1.1.6. Table: divisas

## Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int Auto Increment | ✓ | |
| 2 | nombre | varchar(50) | ✓ | |
| 3 | simbolo | varchar(5) | ✓ | |
| 4 | tasa_cambio | varchar(45) | ✓ | |

| | | | | |
|---|---|---|---|---|
| 5 | estatus | enum | ✓ | Default: 'Activa' |
| 6 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |

## Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| divisas_AFTER_INSERT | AFTER | ✓ | | |
| divisas_AFTER_UPDATE | AFTER | | ✓ | |
| divisas_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.7. Table: metodos_pago

## Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |
| 2 | usuario_id | int | ✓ | Index: fk_usuarios3_idx |
| 3 | numero_tarjeta | varchar(16) | ✓ | |
| 4 | tipo_red_bancaria | enum | ✓ | |
| 5 | tipo_cuenta | enum | ✓ | |
| 6 | titular | varchar(100) | ✓ | |
| 7 | fecha_expiración | char(5) | ✓ | |
| 8 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |
| 9 | fecha_actualizacion | datetime | | |
| 10 | estatus | enum | ✓ | Default: 'Vigente' |

## Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_usuarios3_idx | NORMAL | BTREE | usuario_id |

## Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_usuarios3 | usuario_id | ecommerce_230173.usuarios | persona_fisica_id | RESTRICT | RESTRICT |

## Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| metodos_pago_AFTER_INSERT | AFTER | ✓ | | |
| metodos_pago_AFTER_UPDATE | AFTER | | ✓ | |
| metodos_pago_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.8. Table: pedidos

## Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |
| 2 | carrito_id | int | ✓ | Index: fk_carrito2_idx<br>idx_pedidos_carrito |
| 3 | total_productos | int | ✓ | Default: 0 |
| 4 | importe_total | decimal(10, 2) | ✓ | Default: 0.00 |
| 5 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP<br>Index: idx_pedidos_fecha |
| 6 | fecha_actualizacion | datetime | | |
| 7 | estatus | enum | ✓ | |
| 8 | aprobacion | bit(1) | ✓ | |

## Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_carrito2_idx | NORMAL | BTREE | carrito_id |
| idx_pedidos_carrito | NORMAL | BTREE | carrito_id |
| idx_pedidos_fecha | NORMAL | BTREE | fecha_registro |

## Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_carrito2 | carrito_id | ecommerce_230173.carrito | id | RESTRICT | RESTRICT |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| pedidos_AFTER_INSERT | AFTER | ✓ | | |
| pedidos_AFTER_UPDATE | AFTER | | ✓ | |
| pedidos_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.9. Table: persona_fisica

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id | int | ✓ | Index: id_UNIQUE<br>fk_persona_fisica |
| 2 | titulo_cortesia | varchar(20) | | |
| 3 | nombre | varchar(60) | ✓ | |
| 4 | primer_apellido | varchar(60) | ✓ | |
| 5 | segundo_apellido | varchar(60) | | |
| 6 | fecha_nacimiento | date | ✓ | |
| 7 | genero | enum | | |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| id_UNIQUE | UNIQUE | BTREE | id |
| fk_persona_fisica | NORMAL | BTREE | id |

### Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_persona_fisica | id | ecommerce_230173.personas | id | RESTRICT | RESTRICT |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| persona_fisica_AFTER_INSERT | AFTER | ✓ | | |
| persona_fisica_AFTER_UPDATE | AFTER | | ✓ | |
| persona_fisica_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.10. Table: persona_moral

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id | int | ✓ | Index: id_UNIQUE fk_persona_moral |
| 2 | razon_social | varchar(200) | ✓ | Index: razon_social_UNIQUE |
| 3 | tipo | varchar(100) | ✓ | |
| 4 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |
| 5 | fecha_creación | date | ✓ | |
| 6 | estatus | enum | ✓ | Default: 'Activa' |
| 7 | responsabilidad | varchar(100) | | |
| 8 | capacidad_juridica | varchar(100) | | |
| 9 | patrimonio | varchar(100) | | |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| id_UNIQUE | UNIQUE | BTREE | id |
| razon_social_UNIQUE | UNIQUE | BTREE | razon_social |
| fk_persona_moral | NORMAL | BTREE | id |

### Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_persona_moral | id | ecommerce_230173.personas | id | RESTRICT | RESTRICT |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| persona_moral_AFTER_INSERT | AFTER | ✓ | | |
| persona_moral_AFTER_UPDATE | AFTER | | ✓ | |
| persona_moral_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.11. Table: personas

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |
| 2 | tipo | enum | ✓ | Default: 'Fisica' |
| 3 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |
| 4 | estatus | enum | ✓ | Default: 'Activo' |
| 5 | rfc | varchar(13) | | |
| 6 | codigo_pais | char(2) | ✓ | |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| personas_AFTER_INSERT | AFTER | ✓ | | |
| personas_AFTER_UPDATE | AFTER | | ✓ | |
| personas_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.12. Table: productos

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | |
| 2 | nombre | varchar(150) | ✓ | |
| 3 | descripcion | text | | |
| 4 | marca | varchar(100) | ✓ | |
| 5 | modelo | varchar(100) | | |
| 6 | precio_menudeo | decimal(10, 2) | ✓ | Default: 0.00 |
| 7 | precio_mayoreo | decimal(10, 2) | | |
| 8 | tipo | enum | ✓ | Default: 'No Perecedero' |
| 9 | estatus | enum | ✓ | Default: 'Activo' |
| 10 | codigo_barras | varchar(40) | ✓ | Index: codigo_barras |
| 11 | sku | varchar(40) | ✓ | Index: sku |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| codigo_barras | UNIQUE | BTREE | codigo_barras |
| sku | UNIQUE | BTREE | sku |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| productos_AFTER_INSERT | AFTER | ✓ | | |
| productos_AFTER_UPDATE | AFTER | | ✓ | |
| productos_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.13. Table: productos_categorias

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | categoria_id 🔑 | int | ✓ | |
| 2 | producto_id 🔑 | int | ✓ | Index: fk_producto |
| 3 | estatus | enum | | Default: 'Activo' |
| 4 | fecha_registro | datetime | | Default: CURRENT_TIMESTAMP |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_producto | NORMAL | BTREE | producto_id |

### Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_categoria | categoria_id | ecommerce_230173.categorias | id | RESTRICT | RESTRICT |
| fk_producto | producto_id | ecommerce_230173.productos | id | RESTRICT | RESTRICT |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| productos_categorias_AFTER_INSERT | AFTER | ✓ | | |
| productos_categorias_AFTER_UPDATE | AFTER | | ✓ | |
| productos_categorias_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.14. Table: proveedores

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int Auto Increment | ✓ | |
| 2 | persona_id | int | ✓ | Index: fk_proveedor_idx |
| 3 | tipo | enum | ✓ | Default: 'Menudeo' |
| 4 | estatus | enum | ✓ | Default: 'Activo' |
| 5 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |
| 6 | total_productos | int | ✓ | Default: 0 |
| 7 | monto_acumulado | float | ✓ | Default: 0 |
| 8 | fecha_ultima_compra | datetime | | |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_proveedor_idx | NORMAL | BTREE | persona_id |

### Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_proveedor | persona_id | ecommerce_230173.personas | id | RESTRICT | RESTRICT |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| proveedores_AFTER_INSERT | AFTER | ✓ | | |

| | | | | |
|---|---|---|---|---|
| proveedores_AFTER_UPDATE | AFTER | | ✓ | |
| proveedores_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.15. Table: sesiones

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int<br>Auto Increment | ✓ | Index: idx_sesiones_usuario |
| 2 | usuario_id | int | ✓ | Index: fk_usuarios1_idx |
| 3 | origen | enum | ✓ | Default: 'Plataforma' |
| 4 | tipo_dispositivo | enum | ✓ | |
| 5 | sistema_operativo | varchar(50) | ✓ | |
| 6 | codigo_pais | char(2) | ✓ | |
| 7 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP<br>Index: idx_sesiones_fechas |
| 8 | fecha_inicio | datetime | | Index: idx_sesiones_fechas |
| 9 | fecha_fin | datetime | | Index: idx_sesiones_fechas |
| 10 | estatus | enum | ✓ | Default: 'Activa' |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_usuarios1_idx | NORMAL | BTREE | usuario_id |
| idx_sesiones_usuario | NORMAL | BTREE | id |
| idx_sesiones_fechas | NORMAL | BTREE | fecha_registro, fecha_inicio, fecha_fin |

### Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_usuarios1 | usuario_id | ecommerce_230173.usuarios | persona_fisica_id | RESTRICT | RESTRICT |

### Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| sesiones_AFTER_INSERT | AFTER | ✓ | | |
| sesiones_AFTER_UPDATE | AFTER | | ✓ | |
| sesiones_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.16. Table: transacciones_financieras

### Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | id 🔑 | int(10) Auto Increment | ✓ | Index: idx_transacciones_pedido |
| 2 | tipo | enum | ✓ | |
| 3 | origen | enum | ✓ | |
| 4 | clave_interbancaria | char(18) | | |
| 5 | numero_convenio | varchar(50) | | |
| 6 | monto | decimal(10, 2) | ✓ | Default: 0.00 |
| 7 | metodo_pago_id | int(10) | | Index: fk_metodopago_idx |
| 8 | estatus | enum | | |
| 9 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP Index: idx_transacciones_fecha |
| 10 | fecha_autorizacion | datetime | | |

### Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| fk_metodopago_idx | NORMAL | BTREE | metodo_pago_id |
| idx_transacciones_pedido | NORMAL | BTREE | id |
| idx_transacciones_fecha | NORMAL | BTREE | fecha_registro |

### Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_metodopago | metodo_pago_id | ecommerce_230173.metodos_pago | id | RESTRICT | RESTRICT |

## Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| transacciones_financieras_AFTER_INSERT | AFTER | ✓ | | |
| transacciones_financieras_AFTER_UPDATE | AFTER | | ✓ | |
| transacciones_financieras_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.1.17. Table: usuarios

## Fields

| Pos | Name | Type | Not Null | Others |
|---|---|---|---|---|
| 1 | persona_fisica_id | int | ✓ | Index: persona_fisica_id_UNIQUE fk_persona_fisica2 |
| 2 | nickname | varchar(80) | ✓ | Index: nickname_UNIQUE idx_usuarios_nickname |
| 3 | contrasena | varchar(300) | ✓ | |
| 4 | fecha_registro | datetime | ✓ | Default: CURRENT_TIMESTAMP |
| 5 | fecha_ingreso | datetime | | |
| 6 | estatus | enum | ✓ | Default: 'Activo' |

## Indexes

| Name | Type | Method | Fields |
|---|---|---|---|
| nickname_UNIQUE | UNIQUE | BTREE | nickname |
| persona_fisica_id_UNIQUE | UNIQUE | BTREE | persona_fisica_id |
| fk_persona_fisica2 | NORMAL | BTREE | persona_fisica_id |
| idx_usuarios_nickname | NORMAL | BTREE | nickname |

## Foreign Keys

| Name | Fields | Referenced Table | Referenced Fields | On Delete | On Update |
|---|---|---|---|---|---|
| fk_persona_fisica2 | persona_fisica_id | ecommerce_230173.persona_fisica | id | RESTRICT | RESTRICT |

## Triggers

| Name | Time | Insert | Update | Delete |
|---|---|---|---|---|
| usuarios_AFTER_INSERT | AFTER | ✓ | | |
| usuarios_AFTER_UPDATE | AFTER | | ✓ | |
| usuarios_AFTER_DELETE | AFTER | | | ✓ |

## 2.1.2. Views

### 2.1.2.1. View: atractividad_productos_por_anio

#### Fields

| Name | Type | Null |
|---|---|---|
| producto | text | |
| anio | bigint | ✓ |
| total | bigint | ✓ |

### 2.1.2.2. View: clientes_proveedores

#### Fields

| Name | Type | Null |
|---|---|---|
| Rol | varchar(9) | ✓ |
| tipo | varchar(6) | ✓ |
| persona_id | int | ✓ |
| estudiante | varchar(100) | |
| Nombre | varchar(203) | |

### 2.1.2.3. View: clientes_proveedores_duplicados

#### Fields

| Name | Type | Null |
|---|---|---|
| estudiante | varchar(100) | |
| tipo_persona | enum | ✓ |
| ID | int | ✓ |
| nombre | varchar(203) | |
| esCliente | varchar(2) | ✓ |
| esProveedor | varchar(2) | ✓ |

### 2.1.2.4. View: muestra_triggers

#### Fields

| Name | Type | Null |
|---|---|---|
| TRIGGER_NAME | varchar(64) | ✓ |
| EVENT_MANIPULATION | enum | ✓ |
| EVENT_OBJECT_TABLE | varchar(64) | ✓ |
| ACTION_TIMING | enum | ✓ |
| ACTION_STATEMENT | longtext | ✓ |

### 2.1.2.5. View: nacionalidad_genero

#### Fields

| Name | Type | Null |
|---|---|---|
| nacionalidad | char(2) | ✓ |
| genero | enum | |
| cantidad | bigint | ✓ |

### 2.1.2.6. View: personas_anio_max_registro

#### Fields

| Name | Type | Null |
|---|---|---|
| anio_registro | int | |
| cantidad_personas | bigint | ✓ |

### 2.1.2.7. View: personas_año_regsitro

#### Fields

| Name | Type | Null |
|---|---|---|
| anio_registro | int | |
| cantidad_personas | bigint | ✓ |

### 2.1.2.8. View: productos_atractivos

#### Fields

| Name | Type | Null |
|---|---|---|
| producto_id | int | ✓ |
| nombre | text | |
| total_agregaciones | bigint | ✓ |

### 2.1.2.9. View: productos_sin_origen

#### Fields

| Name | Type | Null |
|---|---|---|
| id_producto | int | ✓ |
| producto | varchar(150) | ✓ |
| estudiante_registro | varchar(100) | |

### 2.1.2.10. View: registros_por_estudiante

#### Fields

| Name | Type | Null |
|---|---|---|
| Estudiante | varchar(100) | ✓ |
| tabla | varchar(100) | ✓ |
| operacion | enum | ✓ |
| Total_Registros | bigint | ✓ |

### 2.1.2.11. View: usuarios_rango_edad

#### Fields

| Name | Type | Null |
|---|---|---|
| segmento_edad | varchar(15) | ✓ |
| genero | enum | |
| total_personas | bigint | ✓ |

### 2.1.2.12. View: vista_origen_pagos

#### Fields

| Name | Type | Null |
|---|---|---|
| metodo_pago | enum | ✓ |
| total_usuarios | bigint | ✓ |

## 2.1.2.13. View: vista_proceso_compra_completo

### Fields

| Name | Type | Null |
|---|---|---|
| nombre_completo_comprador | varchar(182) | |
| fecha_sesion | datetime | |
| duracion_sesion_minutos | bigint | |
| fecha_creacion_carrito | datetime | ✓ |
| total_productos_carrito | int | ✓ |
| monto_estimado_carrito | double | ✓ |
| fecha_pedido | datetime | |
| total_productos_pedido | int | |
| importe_total_pedido | decimal(10, 2) | |
| estatus_pedido | enum | |
| fecha_pago | datetime | |
| importe_pago | decimal(10, 2) | |
| origen_pago | enum | |
| tiempo_total_proceso_minutos | bigint | |

## 2.1.3. Functions

### 2.1.3.1. Function: actualizar_fechas_personas

Function Type: Procedure

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

### 2.1.3.2. Function: calcula_edad

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | v_fecha_nacimiento | Date |

### 2.1.3.3. Function: crear_usuarios

Function Type: Procedure

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

### 2.1.3.4. Function: duracion_sesion

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | p_sesion_id | INT |

### 2.1.3.5. Function: elige_cantidad

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

### 2.1.3.6. Function: elige_dispositivo

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

### 2.1.3.7. Function: elige_divisa

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

### 2.1.3.8. Function: elige_pais

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

### 2.1.3.9. Function: elige_plataforma

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

### 2.1.3.10. Function: elige_producto

Function Type: Function

#### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | v_categoria | varchar |

### 2.1.3.11. Function: elige_sesion

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

## 2.1.3.12. Function: elige_sistema_operativo

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | v_dispositivo | varchar |

## 2.1.3.13. Function: elige_usuario

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

## 2.1.3.14. Function: elimina_caracteres_especiales

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | p_texto | text |

## 2.1.3.15. Function: eliminar_caracteres_especiales

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | p_texto | VARCHAR |

## 2.1.3.16. Function: fn_tiempo_compra

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | fecha_base | DATETIME |

## 2.1.3.17. Function: genera_fechahora_lapso

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | fecha_base | DATETIME |
| IN | tipo | VARCHAR |

## 2.1.3.18. Function: genera_fechahora_rangos

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | v_min | DATETIME |
| IN | v_max | DATETIME |

## 2.1.3.19. Function: genera_numero_aleatorio_rangos

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | v_limite_inferior | int |
| IN | v_limite_superior | int |

## 2.1.3.20. Function: limpiar_bd

Function Type: Procedure

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | reset_password | varchar |

## 2.1.3.21. Function: saluda

Function Type: Function

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| (No parameters) | | |

## 2.1.3.22. Function: simula_carrito

Function Type: Procedure

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | p_cuantos | INT |
| IN | p_total_productos | INT |
| IN | p_categoria | VARCHAR |

## 2.1.3.23. Function: simula_carrito2

Function Type: Procedure

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | p_cuantos | INT |
| IN | p_total_productos | INT |
| IN | p_categoria | VARCHAR |
| IN | p_subcategoria | VARCHAR |

## 2.1.3.24. Function: simula_carritos_samuel

Function Type: Procedure

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | v_cuantos | int |
| IN | v_total_productos | int |
| IN | v_categoria | varchar |

## 2.1.3.25. Function: simula_compras

Function Type: Procedure

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | p_cuantos | INT |
| IN | p_pedido_inicio | INT |
| IN | p_pedido_fin | INT |

## 2.1.3.26. Function: simula_pedidos

Function Type: Procedure

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | p_cuantos | INT |
| IN | p_carrito_inicio | INT |
| IN | p_carrito_fin | INT |
| IN | p_anio | INT |

## 2.1.3.27. Function: simula_sesiones

Function Type: Procedure

### Parameters

| Mode | Name | Data Type |
|---|---|---|
| IN | v_cuantos | INT |